Corrosion-informed learning with TabICL

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Goal

- Create and train TabICL on synthetic samples that match the corrosion dataset.

How:

- Sample from the corrosion dataset and add controlled perturbations to the composition.
- Divide features into **composition, environment, test procedure, material history**, etc.
- Sample features in groups to preserve meaningful relationships.

Problem

- We cannot simply reuse the target of each sampled row:
  - Copying labels would be redundant
  - The features have changed after perturbation
- We need a new way to generate those targets.

Approaches

1. Use **SCM**, but keep the features fixed and only generate the target column.

2. Use **mathematical formulas** based on a priori corrosion knowledge.

1. Approach: SCM

- Start with the sampled corrosion features.
- Keep the features fixed and generate only the target column.
- Sample an **MLP or tree mapping** for each synthetic table.
- Apply the same mapping to all rows in that table.

$$X_{\text{fixed}} \xrightarrow{\text{sampled SCM mapping}} y_{\text{SCM}}$$

SCM = structural causal model · MLP = multilayer perceptron.

2. Approach: Corrosion formulas

Take each synthetic sample's **feature values** and plug them into a **mathematical rule based on corrosion knowledge** to generate a new target.

Example: PREN-N rule, using standardized feature scores

$$y_{\text{formula}} = a \underbrace{P}_{\text{material resistance}} - b \underbrace{E}_{\text{environment aggressiveness}} - c \underbrace{I}_{\text{extra chloride penalty}}$$

Feature values used	Contribution to the generated target
Cr, Mo, W, N	Material resistance: higher resistance raises the target.
Temperature, chloride, pH	Environment: greater aggressiveness lowers the target.
Composition and chloride together	Extra penalty for less-resistant alloys in high chloride.

Fit **a, b, c** on **dev context**; score correlation with real targets on **dev query**. Exclude the final-evaluation set.

2. Approach: Correlations

Formula family	Mean Spearman correlation
PREN-N	0.506
Cr–Mo/W–N synergy	0.534
Threshold / saturation	0.510
PREN-N with improved environment terms	0.551
Mo/W–N acidic-repassivation interaction	0.549
Mn–S penalty	0.571
Coupled breakdown	0.534
Fe/Ni-specific Cr threshold	0.521
Method-aware PREN-N	0.591

Saved EPIT v4 calibration · Mean over five development folds · Eligible Fe / Ni–Cr–Mo rows.
Formula correlations, not final-model evaluation results.

Idea: Mix SCM and formula targets

- Mix SCM targets and formula targets into a **single target used for training**.
- Standardize both target distributions to the same scale before mixing.
- Tune the mixing coefficient in Optuna: `informed_target_mix_weight`.

$$y_{\text{mixed}} = (1 - \alpha)\, z(y_{\text{SCM}}) + \alpha\, z(y_{\text{formula}})$$

- Standardize the mixed target again before training.

α = 0: SCM only · α = 1: formula only. The informed/generic dataset ratio is a separate setting.

Current workflow: Dataset and formulas

- Focus on one specific corrosion task → **Epit**.
- Find a relevant dataset.
- Apply preprocessing to match feature names, units, etc. across datasets.
- Split the dataset into **development data** and an **untouched final-evaluation set**.
  - Further divide development data into dev-context and dev-query folds.
- Develop, optimize, and evaluate target formulas.
  - Optimize on dev context; evaluate on dev query.

Current workflow: Optuna

- Run an Optuna study to find the best hyperparameters for the dataset.
- Use fully informed datasets: **informed_ratio = 1.0**.
- Train each trial for **1,000 steps**.
- Tune:
  - MLP / tree split for SCM.
  - Mixing between SCM targets and formula targets.
  - Feature perturbation strength.
- Evaluate trials on the development folds.

Current workflow: Training and evaluation

- Select the best Optuna trial and keep its tuned hyperparameters.
- Train for **10,000 steps** with different informed/generic dataset ratios:
  - **0.25 · 0.50 · 0.75 · 1.00** informed.
- Select models and checkpoints using development folds, then freeze them.
- Evaluate each frozen model using the **development data as context** and the **untouched final-evaluation set as query**.
- Keep final query rows fixed across comparisons.
- Repeat this process for each new dataset.

Current workflow: Combined model

- Eventually combine two separate **Epit priors** into a single model.
- Sample between the two informed priors and the generic prior, using the ratios selected for each dataset.
- Each prior retains its own tuned hyperparameters.
- Evaluate on a **third Epit dataset**, unused during prior development and training.
  - Use separate labeled-context and held-out query partitions from that dataset.
  - Freeze the model and keep the final queries fixed.
- Eventually expand to other corrosion tasks.

Current workflow: Example mixing ratios

Individually selected configurations:

- Prior 1 → **informed_ratio: 0.50**.
- Prior 2 → **informed_ratio: 1.00**.

Give the two configurations equal weight in the combined model:

Combined training source	Ratio
Informed prior 1	0.25
Informed prior 2	0.50
Generic prior	0.25

General ideas

- Leave the **generic prior untouched** to match original TabICL.
- Do not introduce changes to informed generation that are not strictly relevant, so we can better determine the informed contribution.
- Eventually tune higher-level shared hyperparameters: **learning rate, architecture, and SCM**.
- Introduce additional components, such as a **missingness vector or embedding**.